

MEDHA BHAVANI GIRISH

+91 99725 76345 medhabg@gmail.com [Linkedin](#) [GitHub](#)

Final year undergraduate student in Electrical Engineering at IIT Madras passionate about working at the intersection of signal processing, communication systems and AI/ML.

EDUCATION

Indian Institute of Technology Madras

2023-2027

B.Tech in Electrical Engineering; GPA: 9.30/10

ACHIEVEMENTS

- Recipient of MV Prize awarded to the female student with the **best all-round performance** up to 6 semesters.
- All India Rank 132 in JEE Mains 2023 among 1.16 million candidates; ranked **3rd nationally among girls**.
- All India Rank 1632 in JEE Advanced 2023 among 200,000 candidates.
- SAT Score of 1580/1600 in 2023, achieving 99th percentile.
- All India Rank 14 in Undergraduate Common Entrance Exam for Design (UCEED) 2023 among 14,000 candidates.

SKILLS

- **Languages:** Python, C++, C, MATLAB, SQL, AVR Assembly, ARM Assembly, Verilog, Bash, JavaScript
- **Libraries and Frameworks:** PyTorch, TensorFlow, Keras, Scikit-learn, Optuna, NumPy, SciPy, Pandas
- **ML Techniques:** Deep Learning, CNNs, GNNs, RNNs, LSTMs, Transformers, Attention, Autoencoders, Time-Series Prediction, Reinforcement Learning, Hyperparameter Optimization, Feature Engineering, Regression Analysis
- **Tools:** Git, Linux, Wireshark, Xilinx Vivado, LTspice, Keil µVision, Microchip Studio, EasyEDA, ARM Cortex-M

PUBLICATIONS

M. B. Girish, M. Singh, D. Venkitesh, L. N. Theagarajan, "Attention-assisted deep neural network for nonlinear equalization and denoising in time-stretched photonic ADC," Optics Express, vol. 34, 2026 ([Link](#)) 2026

- Proposing a photonic time-stretch front-end ADC architecture with an **ultra-wideband signal recovery methodology reducing bandwidth requirement by 3x** and agnostic to sampling rate and signal power.
- Developed a **custom deep neural architecture** that integrates an attention-enhanced LSTM and a feed-forward neural network in parallel for suppression of nonlinear distortions induced by the front-end.
- Presented this work at the **Electrical Engineering Research Scholars' Symposium** at IIT Madras along with the PhD and MS scholars as a result of my work being selected as one of the **top undergraduate research projects**.

PROFESSIONAL EXPERIENCE

Texas Instruments Embedded Software: Summer Intern

2026

- Selected as a summer intern in the **Automotive Applications team**, building end-applications for the F29x microcontroller. Worked on flash memory protection, pipeline to load multi-core programs in single-core flash, and secondary bootloader to write data in flash via UART. Presented how to use **TI EdgeAI Studio to train CNNs** for power grid fault detection for EV charging stations taking into account the latency and memory constraints.
- Identified, debugged and fixed defects in production libraries and technical documentation.

Pan-India Inter-IIT Tech Meet 2025: Problem Statement posed by Arista Networks ([Report Link](#))

2025

Adaptive Channel Sensing and Client-Centric Wi-Fi Radio Resource Management

- Developed a multi-armed bandits based algorithm for third Wi-Fi radio for **adaptive channel scheduling** and dwell times based on channel noise and coherence, replacing naive periodic scanning. Achieved **3.99% average improvement** in channel power estimation than periodic scanning when evaluated on correlated I/Q data.
- Integrated MAB-derived channel rankings into a **Wi-Fi interference graph** and channel-allocation objective, enabling channel selection to account for both network-wide interference and local channel quality.
- Built an end-to-end additional-radio sensing pipeline combining **CNN-based interference classification**, MAB-based adaptive scanning, and **CUSUM change detection** for automated channel-switch recommendations.
- Presented algorithm to a panel of judges from Arista networks at Inter-IIT Tech Meet in Patna.

Texas Instruments WiSH Scholar: Embedded Software Track

2025

- Selected as one among 40 students **shortlisted out of 2000+ applicants** across India for month-long mentorship.
- Worked with C2000 microcontrollers implementing a PWM-controlled stepper motor, integrating sensors.

Teaching Assistant - EE1101 Signals and Systems

2025

- Held the position of TA for a class of 160 students. Conducted problem-solving sessions and clarified doubts.

UNDERGRADUATE RESEARCH

Improving Convergence Rate and Accuracy in DSP Algorithms for Adaptive Equalization ([Report Link](#))

2026

Undergraduate Research Project; Guides: Prof. L N Theagarajan, Prof. D Venkitesh

- Implemented and benchmarked adaptive equalization algorithms (Constant Modulus Algorithm with variable momentum, 2nd moment, 2nd order L-BFGS, etc.) for coherent optical communication systems.
- Achieved 8x faster convergence than standard CMA using VM-CMA with comparable accuracy.
- Developed a joint LMS algorithm for simultaneous dispersion and frequency offset compensation, estimating carrier frequency offset within 5% error, eliminating need for separate equalization and 4th order CFO correction.

Undergraduate Research Project; Guides: Prof. L N Theagarajan, Prof. D Venkitesh

- Worked on developing a single compensation system using RL frameworks for correcting different receiver impairments in fiber optical channels to improve efficiency and prevent cascading of errors.
- Formulated polarization and dispersion impairment correction in 2x2 MIMO fibre-optic systems as a Multi-Armed Bandit, using adaptive equalizer coefficient configurations as arms and CMA error as the reward.
- Implemented UCB-based filter tap selection; demonstrated convergence to optimal coefficient configuration among over 500,000+ options in ~1,000 rounds on 100 impaired QPSK symbols. Extended formulation to Q-Learning using experience replay and teacher training from classical CMA; evaluated over 400 episodes, development in progress.

PERSONAL PROJECTS & COURSE PROJECTS**Review of Wi-fi Coverage in Indian workplaces ([Report Link](#))**

2026

- Conducted an indoor WLAN site survey across 32 measurement locations, capturing ~300 IEEE 802.11 beacon frames per location using a WLAN adapter in monitor mode. Analysed PCAP traces to estimate RSSI, and developed an empirical indoor path loss model using linear regression with distance and wall attenuation.
- Estimated propagation parameters including RSSI reference, path loss exponent, and wall attenuation.

Network Traffic Simulator - Communication Networks Course Project ([Link](#))

2026

- Developed a configurable traffic network simulation framework supporting arbitrary topologies, dynamic packet generation, **congestion-aware routing**, and multiple junction types. Implemented dynamic Dijkstra routing with BPR congestion modelling. Analyzed average travel time, throughput, and queue backlog.

Dataset Condensation for Time Series Classification via Dual Domain Matching ([Repository Link](#))

2026

- Implemented the CondTSC time-series dataset condensation algorithm in PyTorch.
- Using dual-domain gradient matching achieved **82.5% accuracy** with only **4% of original training samples**, comparable to full-dataset ML baselines. Conducted cross-architecture and ablation studies, validating transferability across CNN/TCN/LSTM models and demonstrating the role of frequency-domain information.

RELEVANT COURSEWORK

Digital Signal Processing
Information Theory
Estimation Theory
Communication Networks

Communication Systems
RF and Optical Communication
Quantum Computation &
Quantum Information

Convex Optimization
Foundations of Machine Learning
Advanced topics in AI- Data Privacy
Foundations of Probability

EXTRACURRICULARS AND LEADERSHIP**Product Design**

- Led the Product Design Club in the Centre for Innovation at IIT Madras, managing a 3-tier team of 50+ members.
- Founded and developed InstiLite, a mobile app enhancing student engagement by streamlining event management.
- Developed a cost-effective solution for early detection of diabetic neuropathy using quantum dot biosensors. Received a special mention in DiaTech 10X Hackathon conducted by Emory University, Georgia Tech and IITM.

Outreach, Mentoring and Management

- Held the role of India Student Ambassador for Technovation, a California-based program for school girls, addressing the gender gap in tech and entrepreneurship, empowering girls to be problem-solvers. Introduced the program to 200+ girls in 6 schools. Mentored them for a year ideating and implementing social innovation ideas through code.
- Interned at the Times of India as a student journalist- India's most-read English newspaper. Held position of Student Guest Editor for the Times of India NPS HSR Edition.

Art and Dance

- Trained Mohiniyattam dancer, a dance form from Kerala, for 10+ years.
 - Placed 1st in inter-college Art Marathon at Saarang 2024 and Saarang 2025.
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